Macroeconomics in the Sequence Space

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Outline

1. Why Sequence Space?

2. Local Solution
   - 2.1 SSJ
   - 2.2 Theoretical progress after SSJ
   - 2.3 Application

3. Global Solution
   - 3.1 Repeated transition method
   - 3.2 DL in sequence space
   - 3.3 RL in sequence space

4. Reference

Why Sequence Space?

Sequence Space vs. State Space

Motivation

- In heterogeneous-agent (HA) models, the economy's state is the entire cross-sectional distribution — an infinite-dimensional object.
- Sequence-space methods *sidestep* this challenge: they characterize local dynamics around the steady state without ever carrying the distribution as a state variable.
- The key insight: aggregate equilibrium dynamics can be represented in sequence space as mappings from aggregate shock paths to aggregate outcomes, with Jacobians arising as the local linear approximation of these mappings.

Sequence Space vs. State Space

State Space

Definition 1

A state-space representation describes the economy recursively. At date t, a state variable x_t summarizes all payoff-relevant history, and the model evolves as

$$x_{t+1} = f(x_t, \varepsilon_{t+1}), \qquad y_t = g(x_t).$$

Given x_t, the future is independent of the past.

Sequence Space vs. State Space

Sequence Space

Definition 2

A sequence-space representation characterizes equilibrium as a mapping between entire time paths. We seek sequences $\{X_t\}_{t=0}^{\infty}$ such that equilibrium conditions hold at every date:

$$F\left(\{X_t\}_{t=0}^{\infty}, \{Z_t\}_{t=0}^{\infty}\right) = \mathbf{0},$$

where $\{X_t\}$ collects endogenous variables and $\{Z_t\}$ collects exogenous shocks or policy paths.

Sequence Space vs. State Space

Comparison

Intuition

- State space asks: *what is the current state, and how does it transition?*
- Sequence space asks: *given a shock path, what equilibrium outcome path clears all markets?*

Takeaway

When the distribution is the intractable state variable, sequence space replaces a high-dimensional state-space problem with a representation in terms of equilibrium mappings over aggregate paths, making analysis and computation much more tractable.

Local Solution

Steady State

The Standard Incomplete Markets Model

Core of the canonical HA model: the SIM model

$$V_t(s,a) = \max_{c,a'} \; u(c) + \beta \mathbb{E}\left[V_{t+1}(s',a') \mid s\right], \tag{1}$$

$$\text{s.t.} \quad a' + c = (1+r_t)a + y_t(s) \tag{2}$$

$$a' \geq a \tag{3}$$

We basically need 4 steps to solve the steady state:

1. Discretize the state space
2. Solve for steady state policy functions $a'(s,a)$ and $c(s,a)$
3. Solve for steady state distribution $D(s,a)$
4. Aggregate assets and consumption

Steady State

Step 1: Discretization

Asset Grid for a:

- It is common to use a uniform grid.
- Most nonlinearity occurs near the borrowing constraint $\underline{a}$, so more points should be placed close to the constraint.
- Solution: Use a double exponential grid:

$$a_i = \underline{a} + \exp\left(\exp(u_i) - 1\right) - 1$$

  where u_i is uniform on $[0, \bar{u}]$.

Markov Chain for s:

- Approximate an AR(1) process with persistence ρ.
- For $\rho \approx 1$, use Rouwenhorst's method.

Steady State

Step 2: Steady State policies

We'll use two optimality conditions:
Envelope Condition:

$$V_{a,t}(s,a) = (1+r_t)\, u'(c_t(s,a)) \tag{4}$$

First Order Condition:

$$u'(c_t(s,a)) \geq \beta \mathbb{E}\left[V_{a,t+1}\left(s', a'_t(s,a)\right)\right] \tag{5}$$

with equality unless the budget constraint binds.

The key to solving for steady state policies is to use EGM backward iteration. Each iteration proceeds as follows:

$$V_{a,t+1}\left(s', a'\right) \xrightarrow{(5)} c_t^{\text{endo}}\left(s, a'\right) \xrightarrow{\text{interpolate}} a'_t(s,a) \xrightarrow{(1)} c_t(s,a) \xrightarrow{(4)} V_{a,t}(s,a)$$

Steady State

Step 3: Distribution

Goal: Update the next period distribution $D_{t+1}(s, a)$ using the policy function $a'_t(s, a)$.
Issue: The policy $a'_t(s, a)$ generally does not land exactly on the grid points.

Solution: Use lotteries to distribute probability mass. The idea is to map the current mass to adjacent grid points in the next period based on interpolation weights.

Suppose for two grid points a_i, a_{i+1}, the policy yields:

$$a'_t(s, a) = \pi a_i + (1 - \pi) a_{i+1}$$

where $0 \leq \pi \leq 1$. Then, send mass $\pi D_t(s, a)$ to (s', a_i) and $(1 - \pi) D_t(s, a)$ to (s', a_{i+1}).

Key Idea: No simulation is needed! We iteratively update the distribution until it converges.

Steady State

Step 4: Aggregation

For each period t, the aggregate variables are easily obtained from the distribution and policy functions:

$$C_t = \sum_{s,a} c_t(s,a) \cdot D_t(s,a)$$

$$A_t = \sum_{s,a} a'_t(s,a) \cdot D_t(s,a)$$

Computing HA Jacobians: the "fake news" algorithm

Partial Equilibrium Dynamics

Q: Given time-varying $\{r_t\}_{t=0}^T$ or $\{y_t(s)\}_{t=0}^T$, how to get C_t, A_t?

A: Backward and Forward Iteration!

1. From $V_{a,T} = V_{a,\text{ss}}$, iterate backward to get $a'_t(s,a)$, $c_t(s,a)$ for all $t \geq 0$.
2. From $D_0(s,a) = D_{\text{ss}}(s,a)$, iterate forward to get $D_t(s,a)$ from $a'_t(s,a)$.
3. Compute

$$C_t = \sum_{s,a} c_t(s,a) \cdot D_t(s,a), \qquad A_t = \sum_{s,a} a'_t(s,a) \cdot D_t(s,a)$$

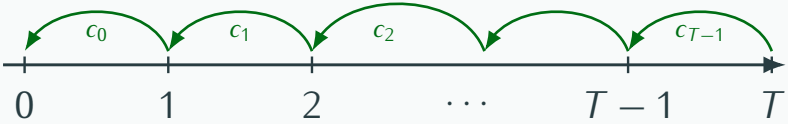
Computing HA Jacobians: the “fake news” algorithm

Partial Equilibrium Dynamics

Computing HA Jacobians: the "fake news" algorithm

Partial Equilibrium Dynamics

Step 1: Backward to get $c_t(s, a)$



Computing HA Jacobians: the "fake news" algorithm

Partial Equilibrium Dynamics

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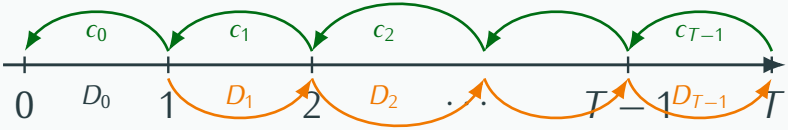
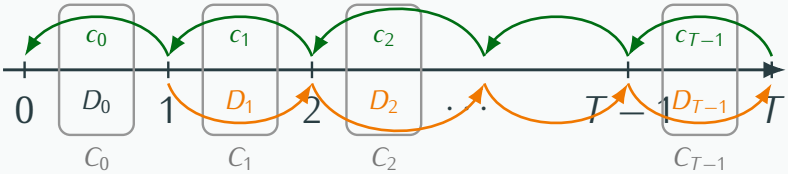


Step 2: Forward to get $D_t(s, a)$

Computing HA Jacobians: the "fake news" algorithm

Partial Equilibrium Dynamics

Step 1: Backward to get $c_t(s, a)$



Step 2: Forward to get $D_t(s, a)$

Step 3: Aggregate to get C_t, A_t

Computing HA Jacobians: the "fake news" algorithm

Sequence-space Jacobians

Denote the impulse response of aggregate consumption C_t to a one-time productivity shock y_s at time s by:

$$\frac{\partial C_t}{\partial y_s}$$

We want the stacked matrix:

$$\mathcal{J}^{C,y} \equiv \left(\frac{\partial C_t}{\partial y_s}\right)_{T\times T} = \left(\mathcal{J}^{C,y}_{ts}\right)_{T\times T}$$

Such a matrix is the sequence-space Jacobian (SSJ).

Computing HA Jacobians: the "fake news" algorithm

How to compute

To compute the SSJ, we truncate to some finite time T (e.g., $T = 300$).
For a generic approach with T columns, one would need T^2 backward iterations and T^2 forward iterations.
Simply loop over all s and t!

Key result: This paper computes the SSJ with only T backward and T forward iterations.

Computing HA Jacobians: the "fake news" algorithm

From $\mathcal{O}(T^2)$ to $\mathcal{O}(T)$: First Row

Denote by $c_t^s(s,a)$ and $a_t^s(s,a)$ the policies at time t in response to a shock at time s.

Key idea 1: Only the time difference $s-t$ until the perturbation matters.

$$c_t^s(s,a) = \begin{cases} c_{ss}(s,a) & \text{if } s < t \\ c_{T-1-(s-t)}^{T-1}(s,a) & \text{if } s \geq t \end{cases}$$

We only compute a single backward iteration with $s = T-1$ to obtain all the $c_t^s(s,a)$.

For aggregate responses:

- $C_0^s = \sum_{s,a} c_0^s(s,a)\, D_{ss}(s,a)$, so we get the first row of the Jacobian: $\mathcal{J}_{0s}^{C,y} = \dfrac{\partial C_0}{\partial y_s}$.
- $D_1^s(s,a)$ is then implied by $c_0^s(s,a)$ for all s.

Computing HA Jacobians: the "fake news" algorithm

From $\mathcal{O}(T^2)$ to $\mathcal{O}(T)$: First Column

- Intuitively, we can continue to compute $D_t^s(s,a)$, then aggregate $c_t^s = c_t^{s'} D_t^s$ for all s, t (Still $O(T^2)$!)
- Observe that to first order,

$$dC_t^s = dc_t^{s'} D_{ss} + c_{ss}' dD_t^s$$

- Consider $s = 0$. For $t \geq 1$, only the second term remains,

$$dC_t^0 = \underbrace{\cancel{dc_t^{0'} D_{ss}}}_{c_t^s(s,a)=c_{ss}(s,a), s<t} + c_{ss}' dD_t^0 = c_{ss}'(\Lambda_{ss}')^{t-1} dD_1^0$$

- Here, Λ_{ss} is the steady-state transition matrix for (s,a), obtained by $c_{ss}(s,a)$ so

$$dC_t^0 = c_{ss}'(\Lambda_{ss}')^{t-1} dD_1^0 \tag{6}$$

- Then we get the first column of the Jacobian: $\mathcal{J}_{t0}^{C,y} = \frac{\partial c_t}{\partial y_0}$ by (6).

Computing HA Jacobians: the "fake news" algorithm

From $\mathcal{O}(T^2)$ to $\mathcal{O}(T)$: Fake news for the rest!

How about $s > 0$? Given policy symmetry, we have $d\mathbf{c}_t^s = d\mathbf{c}_{t-1}^{s-1}$. It follows:

$$dC_t^s - dC_{t-1}^{s-1} = \cancel{\left(d\mathbf{c}_t^s - d\mathbf{c}_{t-1}^{s-1}\right)' \mathbf{D}_{ss}} + \mathbf{c}_{ss}' \left(d\mathbf{D}_t^s - d\mathbf{D}_{t-1}^{s-1}\right)$$

Given $c_t^s = c_{t-1}^{s-1} \implies \Lambda_t^s = \Lambda_{t-1}^{s-1}$, it is then easy to see

$$D_t^s - D_{t-1}^{s-1} = \prod_{j=1}^{t-1} \left(\Lambda_j^{s\prime}\right) \left(\Lambda_0^s - I\right)' D_{ss} = \prod_{j=1}^{t-1} \left(\Lambda_j^{s\prime}\right) \left(D_1^s - D_{ss}\right)$$

Holding $\Lambda_j^s \equiv \Lambda_{ss}$, we have

$$\mathcal{F}_{ts}\, dx \overset{\text{def}}{=} \mathbf{c}_{ss}' \left(d\mathbf{D}_t^s - d\mathbf{D}_{t-1}^{s-1}\right) = \mathbf{c}_{ss}' \left(\Lambda_{ss}'\right)^{t-1} d\mathbf{D}_1^s \equiv \left(\mathcal{J}_{ts}^{C,y} - \mathcal{J}_{t-1,s-1}^{C,y}\right) dx \tag{7}$$

We interpret the $\mathcal{F}_{ts}$ as the "fake news" shock on C_t:

- $t = 0$: news shock that period s shock hits $\Rightarrow d\mathbf{D}_1^s$
- $t = 1$: fake news that there won't be a shock at $s \Rightarrow c_{ss} \Rightarrow \Lambda_{ss}$ afterwards

Computing HA Jacobians: the "fake news" algorithm

From $\mathcal{O}(T^2)$ to $\mathcal{O}(T)$: Whole $\mathcal{J}^{C,y}$

Key Idea 2: Recover $\mathcal{J}^{C,y}$ from 1st row, 1st column and $\mathcal{F}$.
News shock is a sequence of fake news shocks!

$$\mathcal{J}^{C,y} = \begin{pmatrix} \mathcal{J}_{00} & \mathcal{J}_{01} & \mathcal{J}_{02} & \cdots \\ \mathcal{J}_{10} & \mathcal{J}_{00} + \mathcal{F}_{00} & \mathcal{J}_{01} + \mathcal{F}_{01} & \cdots \\ \mathcal{J}_{20} & \mathcal{J}_{10} + \mathcal{F}_{10} & \mathcal{J}_{11} + \mathcal{F}_{11} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

where, by the recursive structure, representative entries are given by:

$$\mathcal{J}_{ts} = \begin{cases} \mathcal{J}_{t0} & \text{if } s = 0 \\ \mathcal{J}_{0s} & \text{if } t = 0 \\ \mathcal{J}_{t-1,s-1} + \mathcal{F}_{t-1,s-1} & \text{if } t, s \geq 1 \end{cases}$$

Computing HA Jacobians: the "fake news" algorithm

Implementation

Define the expectation function:

$$\mathcal{E}_t(s, a) \equiv \mathbb{E}[c_{ss}(s_t, a_t) \mid s_0 = s,\ a_0 = a]$$

that is, the expected consumption at time t for an agent starting at (s, a) today.

This is very easy to compute recursively:

$$\mathcal{E}_0 = c_{ss}, \qquad \mathcal{E}_t = \Lambda_{ss}\mathcal{E}_{t-1}$$

Key Idea 3: The $\mathcal{E}'_t$ are enough to get all $t \geq 1$ rows of $\mathcal{F}$:

- First column ($s = 0$): from (6), $\mathcal{J}_{t0}\, dx = \mathcal{E}'_t\, dD_1^0$
- Other columns ($s > 0$): from (7), $\mathcal{F}_{ts}\, dx = \mathcal{E}'_t\, dD_1^s$
- Result: Eventually only $O(T)$ computation!

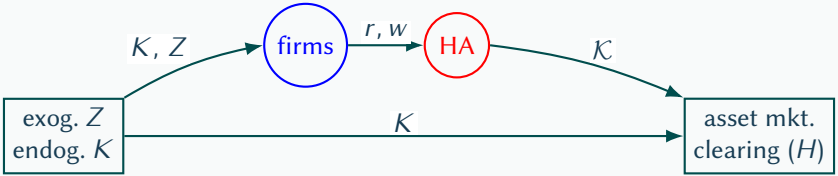
DAG

Krusell-Smith model as a directed acyclic graph (DAG)

Recall that we wrote equilibrium in KS model as system

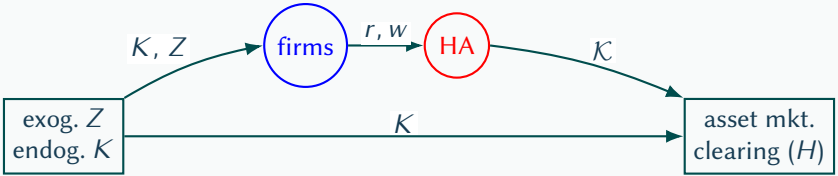
$$H_t(\boldsymbol{K}, \boldsymbol{Z}) \equiv \mathcal{K}_t \left(\left\{ \underbrace{\alpha Z_s K_{s-1}^{\alpha-1} - \delta}_{r_s}, \underbrace{(1-\alpha) Z_s K_{s-1}^{\alpha}}_{w_s} \right\}_{s=0}^{\infty} \right) - K_t = 0$$

where $\mathcal{K}_t \equiv A_t$ is the aggregate capital stock at time t and $\boldsymbol{Z} \equiv \{Z_s\}_{s=0}^{\infty}$ is the time-varying TFP process. This corresponds to a simple graph:



DAG

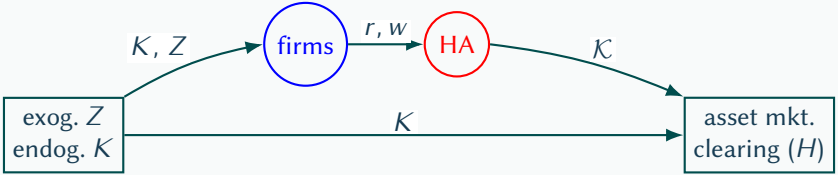
DAGs combining simple and HA blocks



- Each node ("block") takes sequences as inputs and outputs.
- Inputs of later blocks are outputs of earlier blocks.
- The number of endogenous variables ("unknowns") equals the number of equations in H ("targets"); solve to get equilibrium.
- Two kinds of blocks:
  - Simple blocks: Analytical equations directly in aggregates, e.g. $r_t = \alpha Z_t K_{t-1}^{\alpha-1} - \delta$; Jacobians are sparse and easy to obtain.
  - HA blocks: As described previously.

DAG

DAGs combining simple and HA blocks



- Define $\mathbf{J}^{o,i}$ to be the total derivative of o along the DAG with respect to the exogenous or endogenous variable i (distinct from the block Jacobian $\mathcal{J}^{o,i}$).
- Here, for example, $\mathbf{J}^{r,K}$ equals $\mathcal{J}^{r,K}$, etc.: the direct effect is the only effect.
- Then we need the chain rule. For instance:

$$\mathbf{J}^{\mathcal{K},K} = \mathcal{J}^{\mathcal{K},r}\mathbf{J}^{r,K} + \mathcal{J}^{\mathcal{K},w}\mathbf{J}^{w,K}$$

- Finally, we obtain $\mathbf{H}_K$ and $\mathbf{H}_Z$; then $\mathbf{G}^{K,Z} \equiv -\mathbf{H}_K^{-1}\mathbf{H}_Z$ gives the general equilibrium map from Z to K, and all $\mathbf{G}^{o,Z}$ are determined from it.

DAG

General case: solving with forward accumulation

- Initialize $\mathbf{J}^{i,i}$ as the identity for all exogenous or unknown i.
- Evaluate the chain rule following a topological sort:

$$\mathbf{J}^{o,i} = \sum_{m \in I_b} \mathcal{J}^{o,m} \mathbf{J}^{m,i}$$

  on blocks b to get $\mathbf{H}_U = \mathbf{J}^{H,U}$ and $\mathbf{H}_Z = \mathbf{J}^{H,Z}$.
- This is called "forward accumulation" in the algorithmic differentiation literature.
- Compute the general equilibrium mapping: $\mathbf{G}^{U,Z} = -\mathbf{H}_U^{-1}\mathbf{H}_Z$
- Then do forward accumulation again:

$$\mathbf{G}^{o,Z} = \sum_{m \in I_b} \mathcal{J}^{o,m} \mathbf{G}^{m,Z}$$

  to get general equilibrium mappings $\mathbf{G}^{o,Z}$ from shocks dZ to all variables o of interest.
- This gives impulse responses to every shock simultaneously.

Determinacy and Large-Scale Solutions in the Sequence Space

From truncated Jacobians to quasi-Toeplitz operators

Main result

For stationary models, sequence-space Jacobians admit the decomposition

$$J = T(j) + E, \qquad j(z) = \sum_{k=-\infty}^{\infty} j_k z^k,$$

where $T(j)$ is Toeplitz and E is a compact correction.

- Far from date 0, the Jacobian is approximately translation invariant:

$$J_{t,s} \approx j_{t-s}.$$

- E captures missing anticipation: shocks near $t = 0$ are not fully anticipated in the past.
- This applies to stationary heterogeneous-agent blocks & stable linear expectational systems.

Determinacy and Large-Scale Solutions in the Sequence Space

Determinacy in sequence space

To solve equilibrium, we need

$$Jx = y.$$

For block quasi-Toeplitz J, define the matrix-valued symbol $j(z)$ and study $\det j(z)$ on the unit circle.

Winding-number test

$$\operatorname{wind}(\det j) = \operatorname{nonex}(J) - \operatorname{indet}(J).$$

Generically:

$$\begin{cases} \operatorname{wind}(\det j) = 0 & \Rightarrow \text{unique solution}, \\ \operatorname{wind}(\det j) < 0 & \Rightarrow \text{indeterminacy}, \\ \operatorname{wind}(\det j) > 0 & \Rightarrow \text{nonexistence}. \end{cases}$$

- Extends Onatski's Toeplitz test to the quasi-Toeplitz case.
- Equivalent to Blanchard–Kahn in standard linear systems, but works in sequence space.

Determinacy and Large-Scale Solutions in the Sequence Space

Large-scale solution

The idea is to exploit the Toeplitz core instead of directly inverting a huge truncated matrix.

Preconditioned iterative solution

Use an approximate inverse

$$P \approx T(j^{-1}) + \widetilde{E}$$

as a preconditioner, and solve the truncated system iteratively (e.g. GMRES).

- Toeplitz part is cheap to apply/store:

$$T_T(j)x \quad \text{via FFT in } O(T \log T).$$

- Compact correction is compressed in low rank:

$$J_{T,T} \approx T_T(j) + U_{T,r} V'_{T,r}, \qquad r \ll T.$$

- This reduces truncation error and avoids direct $nT \times nT$ inversion.

Determinacy and Large-Scale Solutions in the Sequence Space

Takeaway (Auclert et al., 2025a)

What this adds to SSJ

The paper extends SSJ in two directions:

- theory: existence / uniqueness / determinacy;
- computation: very large sequence-space systems can be solved quickly.

2nd SSJ

From certainty equivalence to certainty correspondence

Known result

At first order, which is the usual certainty equivalence

$$\text{perfect-foresight sequence space} \iff \text{linear stochastic MA}(\infty),$$

Compare:

$$X_t(\{\sigma Z_s\}_{s\geq 0}, D_{ss}) \qquad \text{and} \qquad \widetilde{X}_t = \mathcal{X}_t(\sigma, \epsilon_t, \epsilon_{t-1}, \ldots; \{Z_s\}).$$

Certainty correspondence

$$\text{2nd-order perfect-foresight responses} \iff \text{coefficients of a quadratic MA}(\infty) \text{ for } \widetilde{X}_t.$$

So SSJ is still enough beyond certainty equivalence.

2nd SSJ

What corresponds to what?

Expand the stochastic solution in σ:

$$\widetilde{X}_t = X + \sum_{j\geq 0} x_j \epsilon_{t-j} + \frac{1}{2}\left[\mathcal{X}_{\sigma\sigma}(0)\sigma^2 + \sum_{j,k\geq 0} \mathcal{X}_{\epsilon_{-j}\epsilon_{-k}}(0)\, \epsilon_{t-j}\epsilon_{t-k}\right] + O(\sigma^3).$$

By symmetry in σ,

$$\mathcal{X}_{\sigma\epsilon_{-j}}(0) = 0.$$

Interpretation

- $\{x_j\}$ are the familiar 1st-order perfect-foresight IRFs.
- The quadratic coefficients

$$\mathcal{X}_{\epsilon_{-j}\epsilon_{-k}}(0)$$

  are identified by 2nd-order perfect-foresight responses.

2nd SSJ

Why does this matter?

Rewrite the quadratic MA as

$$\widetilde{X}_t \simeq X + \sum_{j\geq 0} x_j \epsilon_{t-j} + \frac{1}{2}\mathcal{X}_{\sigma\sigma}(0)\sigma^2 + \frac{1}{2}\sum_{j\geq 0} \mathcal{X}_{\epsilon_{-j}\epsilon_{-j}}(0)\epsilon_{t-j}^2 + \sum_{j<k} \mathcal{X}_{\epsilon_{-j}\epsilon_{-k}}(0)\epsilon_{t-j}\epsilon_{t-k}.$$

- Constant term: Anticipated risk.
- ϵ^2 terms: Size dependence.
- $\epsilon_{t-j}\epsilon_{t-k}$ terms: State dependence.

2nd SSJ

Takeaway (Auclert et al., 2025b)

What this adds to SSJ

The paper extends SSJ in two directions:

- Extends SSJ from linear IRFs to nonlinear, state-dependent dynamics in sequence space;
- A practical framework for studying risk, nonlinear propagation, and welfare in rich heterogeneous-agent models.

SSJ in Continuous Time

What changes relative to discrete-time SSJ?

Core comparison

Discrete-time SSJ / FNA:

$$\text{discretize first} \implies \text{differentiate numerically.}$$

Continuous-time SSJ:

$$\text{differentiate in continuous time} \implies \text{discretize later.}$$

- The paper keeps the same sequence-space logic as Auclert et al.: household response → distribution → fake news → Jacobian.
- The main change is only in the first step: instead of numerically perturbing a nonlinear Euler equation, the household response is characterized analytically in continuous time.

SSJ in Continuous Time

Takeaway (Bilal and Goyal, 2025)

- Faster: Jacobians and impulse responses are about 3 times faster to compute.
- Where does the gain come from? From the policy-function step only: linear PDEs replace nonlinear backward solves plus numerical differentiation.
- Why continuous time helps: with a binding borrowing constraint, the first-order condition still holds with equality, so the perturbation is cleaner.

Broader contribution

The paper shows that SSJ / FNA is not tied to discrete time. It extends sequence-space methods naturally to continuous-time HA/HANK models, especially environments with rich heterogeneity and occasionally binding constraints.

Applications of the extended version of SSJ

- Fiscal and monetary policy (Auclert et al., 2025c, Auclert et al., 2024b)
- Het households + het firms (Winberry et al., 2025)
- Behavioural macro (Angeletos et al., 2025, Auclert et al., 2020)
- Life-cycle model (Bardoczy and Velasquez Giraldo, 2025, Bardóczy and Velásquez-Giraldo, 2024)
- Dynamic spatial model (Donald et al., 2025, Fan et al., 2023)
- Production network (Schaab and Tan, 2023)
- Het banks (Bellifemine et al., 2022)
- Optimal policy (Auclert et al., 2024a, McKay and Wolf, 2022)
- Macro labor (Guerreiro et al., 2024, Kekre, 2023)

Global Solution

Repeated transition method

From local SSJ to global sequence space

What is the extension?
Standard SSJ is mainly a local method:

$$\text{1st-order} \;+\; \text{perfect foresight} \;+\; \text{IRFs around a reference point.}$$

RTM extends sequence-space methods to

$$\text{global} \;+\; \text{nonlinear} \;+\; \text{aggregate uncertainty.}$$

- It does not assume perfect foresight.
- It does not require a parametric law of motion for the aggregate distribution.

Repeated transition method

Why "repeated transitions"?

To solve backward at time t, agents need a conditional expectation of continuation values:

$$E_t V_{t+1} = \Pi_{S_t,G} V(\cdot; \Phi_{t+1}, G) + \Pi_{S_t,B} V(\cdot; \Phi_{t+1}, B).$$

RTM's trick

If the simulated path is long enough, a similar aggregate state reappears. Find a period

$$\tilde{t} + 1 = \arg\min_{\tau} \Phi_{\tau}^{(n)} - \Phi_{t+1}^{(n)}$$

with the counterfactual shock realization. Then use the realized value function at $\tilde{t} + 1$ to fill in the missing continuation value.

- Intuition: history repeats itself along an ergodic equilibrium path.
- So expectations are built from repeated transitions, not from an estimated law of motion.

Repeated transition method

Takeaway (Lee, 2025)

- More general than SSJ: can study global nonlinear dynamics, not only local IRFs.
- Avoids law-of-motion misspecification: no need to summarize the distribution with a guessed forecasting rule.

Takeaway

RTM keeps the sequence-space perspective, but turns it into a global stochastic solution method: better suited for nonlinear state dependence, occasionally binding constraints, and models with non-trivial market clearing.

DL in sequence space

From local SSJ to global sequence space

What is the extension?
Standard SSJ is mainly a local method:

$$\text{aggregate state} \xrightarrow{\text{Jacobian}} \text{local IRFs.}$$

This paper instead builds a global approximation to equilibrium in sequence space.

In place of a high-dimensional aggregate state such as (z_t, μ_t), use a truncated shock history

$$z_t^T = (z_{t-T}, z_{t-T+1}, \ldots, z_t),$$

- In ergodic economies, the truncation error vanishes as T grows.
- So the aggregate state is approximated by a sequence of past aggregate shocks, not by moments or histograms of the distribution.

DL in sequence space

How deep learning enters

Parameterize equilibrium objects—policies or prices—with a neural network:

$$\mathcal{N}_\rho(X_t^{\text{seq},T}) \approx \text{equilibrium object at } t.$$

Train the network to satisfy equilibrium conditions along simulated ergodic paths:

$$\rho^* = \arg\min_\rho \frac{1}{N} \sum_{t=1}^{N} \left(\text{equilibrium residual}_t(\rho)\right)^2.$$

- The network input is the truncated history of aggregate shocks.
- The training data come from simulated paths of the economy.

Compared with SSJ

Instead of computing derivatives around a reference point, the method directly learns nonlinear equilibrium mappings over the ergodic set.

DL in sequence space

Takeaway (Azinovic-Yang and Žemlička, 2025)

- Handles richer models globally: large shocks, aggregate uncertainty, and nonlinearities.
- Scales to high-dimensional environments: OLG, heterogeneous households & firms, and life-cycle models with discrete retirement choice.
- Shape-preserving architectures: predicted policies can be forced to be monotone (or concave), which makes EGM usable inside the algorithm.

Takeaway

The paper extends sequence-space methods from fast local perturbation to global nonlinear equilibrium computation, using deep learning to make long shock histories computationally tractable.

RL in sequence space

Agents learn prices (Yang et al., 2025)

The key step is to replace the high-dimensional distribution state by low-dimensional prices:

$$p_t = P^*(g_t, z_t), \qquad \pi_t = \pi(s_t, z_t, p_t).$$

- Households care directly about prices, not the full distribution.
- The algorithm therefore works with prices as state variables.
- This sidesteps the Master equation, where the distribution would enter the Bellman equation.

Reference

Reference (1)

Achdou, Y., Han, J., Lasry, J.-M., Lions, P.-L., and Moll, B. (2022). Income and wealth distribution in macroeconomics: A continuous-time approach. *The review of economic studies*, 89(1):45–86.

Aiyagari, S. R. (1994). Uninsured idiosyncratic risk and aggregate saving. *The Quarterly Journal of Economics*, 109(3):659–684.

Angeletos, G.-M., Guerreiro, J., and Zhang, D. R. (2025). From rank to hank, without fire. Technical report, National Bureau of Economic Research.

Auclert, A., Bardóczy, B., Rognlie, M., and Straub, L. (2021). Using the sequence-space jacobian to solve and estimate heterogeneous-agent models. *Econometrica*, 89(5):2375–2408.

Auclert, A., Cai, M., Rognlie, M., and Straub, L. (2024a). Optimal long-run fiscal policy with heterogeneous agents. *Unpublished Manuscript, Harvard University*.

Auclert, A., Majic, E., Rognlie, M., and Straub, L. (2025a). Determinacy and large-scale solutions in the sequence space. *Journal of Political Economy: Macroeconomics*.

Auclert, A., Rigato, R., Rognlie, M., and Straub, L. (2025b). Beyond certainty equivalence: Second-order dynamics in the sequence space. Presentation slides.

Reference (2)

Auclert, A., Rognlie, M., and Straub, L. (2020). Micro jumps, macro humps: Monetary policy and business cycles in an estimated hank model. Technical report, National Bureau of Economic Research.

Auclert, A., Rognlie, M., and Straub, L. (2024b). The intertemporal keynesian cross. *Journal of Political Economy*, 132(12):4068–4121.

Auclert, A., Rognlie, M., and Straub, L. (2025c). Fiscal and monetary policy with heterogeneous agents. *Annual Review of Economics*, 17.

Azinovic-Yang, M. and Žemlička, J. (2025). Deep learning in the sequence space. *arXiv preprint arXiv:2509.13623.*

Bardóczy, B. and Velásquez-Giraldo, M. (2024). Hank comes of age: Monetary policy with heterogeneous overlapping generations.

Bardoczy, B. and Velasquez Giraldo, M. (2025). Sequence-space jacobians of life cycle models. *Sequence-Space Jacobians of Life Cycle Models (May 29, 2025).*

Bellifemine, M., Jamilov, R., and Monacelli, T. (2022). Hbank: Monetary policy with heterogeneous banks.

Reference (3)

Bilal, A. and Goyal, S. (2025). Some pleasant sequence-space arithmetic in continuous time. Technical report, National Bureau of Economic Research.

Boppart, T., Krusell, P., and Mitman, K. (2018). Exploiting mit shocks in heterogeneous-agent economies: the impulse response as a numerical derivative. *Journal of Economic Dynamics and Control*, 89:68–92.

Donald, E., Fukui, M., and Miyauchi, Y. (2025). Optimal dynamic spatial policy. Technical report, National Bureau of Economic Research.

Fan, J., Hong, S., and Parro, F. (2023). Learning and expectations in dynamic spatial economies. Technical report, National Bureau of Economic Research.

Guerreiro, J., Hazell, J., Lian, C., and Patterson, C. (2024). Why do workers dislike inflation?: Wage erosion and conflict costs.

Kekre, R. (2023). Unemployment insurance in macroeconomic stabilization. *Review of Economic Studies*, 90(5):2439–2480.

Krusell, P. and Smith, Jr, A. A. (1998). Income and wealth heterogeneity in the macroeconomy. *Journal of political Economy*, 106(5):867–896.

Reference (4)

Lee, H. (2025). Global nonlinear solutions in sequence space and the generalized transition function. Technical report, Working paper.

McKay, A. and Wolf, C. (2022). Optimal policy rules in hank. *Manuscript, January*, pages 121–144.

Moll, B. (2025). The trouble with rational expectations in heterogeneous agent models: A challenge for macroeconomics. *The Economic Journal*, page ueaf104.

Schaab, A. and Tan, S. Y. (2023). Monetary and fiscal policy according to hank-io. *Unpublished Manuscript.*

Winberry, T., Auclert, A., Rognlie, M., and Straub, L. (2025). New keynesian economics with household and firm heterogeneity. Technical report, National Bureau of Economic Research.

Yang, Y., Wang, C., Schaab, A., and Moll, B. (2025). Structural reinforcement learning for heterogeneous agent macroeconomics. *arXiv preprint arXiv:2512.18892.*